TSM 099

Homework 8

Lines, slopes, graphs, and systems

Topics 30-34 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

Representing lines

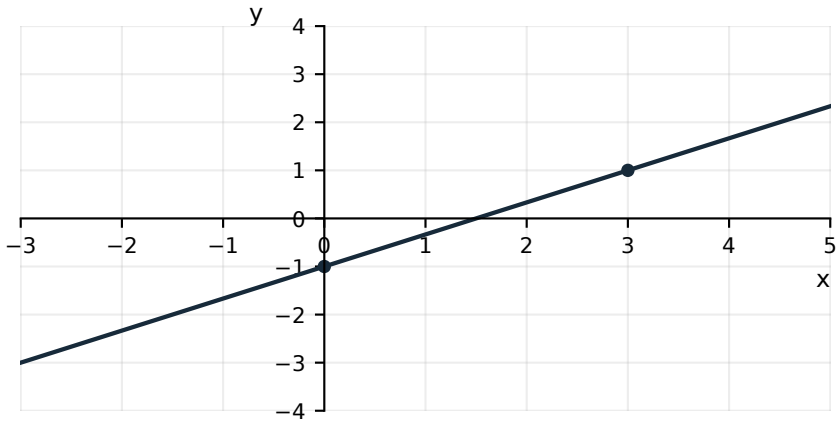
1. Graph by intercepts and check a third point.

$3x + 2y = 6$

2. State the slope and y-intercept, then graph using the slope.

$y = -\frac{2}{3}x + 1$

3. Write the equation of the graphed line in slope-intercept form. Use the marked lattice points.



4. Find the slope through (-2, 5) and (4, -1), then write the line equation in slope-intercept form.

5. Find the slope through (2, -3) and (2, 4). Write the line equation. Compare it with the line through (-1, 4) and (3, 4).

Intersections and algebraic methods

6. Graph the system, read its intersection, and check both equations.

$y = x - 1, \qquad y = -2x + 5$

7. Solve by substitution and check.

$y = 3x - 4, \qquad 2x + y = 11$

8. Solve by elimination and check.

$2x + 3y = 7, \qquad 3x - 2y = 4$

9. Classify and solve each system. Explain the graph relationship.

a) $2x - y = 3, \quad 4x - 2y = 6$

b) $2x - y = 3, \quad 4x - 2y = 8$

10. Find the line through (1, -2) parallel to y = 3x + 4. Check that the point lies on your line and explain why the slopes agree.

11. A student says a negative slope means every point on the line has a negative y-coordinate. Use y = -x + 3 to disprove the claim and explain what the slope means.